

Review log

Errors found and fixed while producing `MODEL.md`, and what was checked. Kept separate so the report stays clean.
2026-08-20/21.

1. Errors in my own work, and how each was caught

#	error	caught by	fix
1	Wrote $V_{\text{mut}}(\mathbf{r}_{\text{mut}}^e) = 0$. The mutant Hamiltonian keeps the stress and relaxation terms; what is zero is V_{ref} , a <i>different</i> Hamiltonian.	Julian	§1 now writes all three Hamiltonians explicitly as equations (2)–(4).
2	Used a synthetic protein (a noisy helix, then a random globule) when <code>pdb_2acy_A</code> was in the package.	Julian	Everything re-run on 2acy chain A.
3	Reimplemented force and linear response instead of reading <code>R/penm.R</code> — which is why I missed that <code>penm</code> already linearises ($f_{ij} = -k_{ij}\delta l_{ij}$).	Julian	Catalogue built on <code>penm</code> ’s own functions. But see §4: independence was right for sclfenm and frustration, which are the things under investigation; it was wrong for the settled parts.
4	Confused <i>response</i> (marginal over mutated sites) with <i>influence</i> (marginal over response sites), and plotted the second labelled as the first.	Julian	§7 rewritten around the response matrix R_{ij} and its two marginals.
5	Claimed “prestress breaks rotational invariance”. Nonsense — l_{ij} and d_{ij} are scalars.	Julian	The three spurious modes came from evaluating the Hessian at the non-stationary LFENM structure ($ F = 0.61$). At the true minimum: exactly 6 zero modes.
6	Confused two meanings of β (Boltzmann vs selection strength).	Julian	$\beta = 1/kT$ throughout; selection is ν .

#	error	caught by	fix
7	Wrote hardcoded conclusions into script output, printed above numbers that contradicted them. Three times: “SYMMETRIC by construction” above 0.49, 0.996, \$-0.13.	Julian	Those scripts deleted. Replacements print only what they measure.
8	RMSIP computed over <i>all</i> modes is identically 1 — two random orthonormal bases give 1. It measured nothing.	code audit	Now over the 10 softest modes: 1.000 for lfenm (frozen spectrum), 0.9937 for sclfenm.
9	§8’s “exact for any state function” claim. Exact only in a <i>fixed</i> background; used mid-trajectory it defines a surrogate V_{cat} .	physics referee	§8 now distinguishes V_{cat} from V , and reports the per-move error.
10	§8(c) was circular: V_{cat} is separable and the proposal symmetric, so a correct sampler <i>must</i> reach the product measure.	physics referee	Now labelled a unit test of the sampler; the honest version (chain on V_{cat} , evaluate exact V) added.
11	“The error does not grow with trajectory length” — false.	physics referee	Measured uniformly: 9 % → 48 % (energy), 0.7 % → 17 % (structure), 5 → 40 substitutions.
12	The $\beta \rightarrow \nu$ rename silently broke <code>fig4_traj.R</code> : it errored on its own default, and passed <code>beta=</code> where the Metropolis rule reads <code>nu=</code> , so two “different” runs were identical.	code audit	Fixed and regenerated. Values in the report were right; the script had rotted around them.
13	Six tests that could not fail (beyond the three §12 already admits).	code audit	I2 rebuilds mutants independently; I3 reports a distribution instead of a forced 50 %; the saddle check reads the raw spectrum by rank.
14	Unseeded scripts (<code>run_frustration.R</code> , <code>fig3_dv_positive.R</code>) produced different numbers every run.	code audit	Seeded; report synced.
15	Stale numbers throughout: values quoted from one script, attributed to another.	code audit, editor	Each traced to its producing script and corrected.

#	error	caught by	fix
16	A phantom figure panel: text cited “Panel (c)” of a two-panel figure.	editor	The number lives in fig 5(c); text corrected.
17	Algebra error in the projector proof: as written, $f = -B\sqrt{k}u$ scales as $k^{3/2}\delta l$.	editor	Corrected, with B defined concretely.
18	Symbols used before definition (K, C, e_{ij} , B); f meaning three different things in one paragraph.	editor	Symbol table extended; Step 2 rewritten.

The recurring failure mode is #7 and #8 and #10: producing a check that cannot report anything but success. Three distinct instances. Worth naming because it is worse than an ordinary error — it manufactures confirmation.

2. Reviewers

Three, all with the code and able to run it.

Physics referee (twice). First round: found the projector proof I had missed (a cleaner result than my numerical demonstration), and that my mechanism for keepnet was contradicted by my own output. Second round: found errors 9, 10, 11 above, and supplied two physics gaps — where relaxedness enters the projector proof ($K = BkB^T$ holds only for a relaxed network: verified, exact when relaxed, off by 0.23/12.9 when frustrated), and why exactly three modes are spurious off-stationarity (translations are annihilated identically; rotations only via $F = 0$).

Code auditor (twice). Verified every number against the script that produces it. Found errors 8, 12, 13, 14, 15.

Editor (once). Found errors 16, 17, 18, plus structural problems: the system description arriving in §7 after six sections of numbers drawn from it, headings that stated topics rather than findings, and five passages of draft archaeology that were cut.

3. Checks that exist, and which can fail

suite	count	can fail?
validate.R V1–V5	5	V1, V2 live; V3, V5 and half of V4 cannot fail — stated as a defect in §12
derivation.R	11	all live: each compares an analytic step against an independent numerical evaluation
test_identities.R	4	I2 and I4 live after the fix; I1 is a genuine linearity test
test_can_fail.R	2 sabotages	demonstrates V1 and V2 go red when the code is broken
saddle check (§9)	1	proved it fires: at $l = 1.5d$ it detects a saddle

4. Two implementations, and whether they agree

The scripts split by when they were written:

- §§1–7 use a reimplementaion (`enm_core.R`, `sclfenm.R`, `applications.R`) written before I was told to read `R/penm.R`.
- §§8–10 use `penm`'s own functions (`set_enm`, `get_mutant_site`, `calculate_force`, `calculate_dxyz`, `ddg_dv`) via `penm_catalog.R` and `frustration.R`.

They were never checked against each other until asked. They agree (`test_two_impls.R`, $\text{ANM} / d_{\text{max}} = 10.5 / 2\text{acy}$):

quantity	agreement
contact set	identical (960 edges, same pairs)
rest lengths l_{ij}	exact
Hessian	3.6×10^{-15} (scale 12.86)
spectrum (288 modes)	1.1×10^{-14}
force from the same δl	1.3×10^{-15}

So the report's numbers are consistent across the split. Two things to note anyway:

1. The duplication is a defect for the settled parts, and correct for the questioned ones. ANM construction, force and linear response are not in doubt — reimplementing those was waste, and it cost the linearisation insight. But `sclfenm` and `frustrated` are precisely what the package flags as unchecked (`stopifnot(!frustrated)`, skipped tests), so building those independently is the point of the exercise, not a mistake. Having both is what made §11 possible: the independent implementation is what the package code can be measured *against*.
2. The mutation model is not `penm`'s default. `enm_core.R` perturbs *every* contact of a site; `penm`'s `generate_delta_lij` honours `mut_sd_min` (default 2, which excludes backbone neighbours). For §§8–10 I passed `mut_sd_min = 1L` to match §§1–7. So the whole report uses `mut_sd_min = 1`, not the package default.

5. Why this took as long as it did

Mostly rework caused by errors 1–7, each of which invalidated a round of results and required re-deriving from scratch. The reviewing was fast; the recovery was not. Concretely:

- Building a synthetic protein instead of using `pdb_2acy_A` cost one full cycle of figures and numbers, all of which had to be regenerated.
- Reimplementing `penm`'s force and response cost the linearisation insight, which is the foundation of the catalogue model — the central result. Reading `R/penm.R` first would have saved the largest single detour.
- The three hardcoded-conclusion scripts each produced a wrong answer that had to be found and undone.

What would have prevented most of it: read the existing implementation before writing any of my own, and use the package's own test data. The nuance learned late (§4): reuse the package for what is settled, reimplement only what is under investigation — and say which is which from the start.